

MARVIN GÜLHAN

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EDUCATION

University of Bonn

M.Sc. Computer Science · Focus: AI & Algorithms · Current grade: 1.1 (1.0 = best)
Incoming exchange semester: Universidad Politécnica de Madrid, Spain

Bonn, Germany

Oct. 2025 – Mar. 2028
Feb. 2027 – Jun. 2027

University of Bonn

B.Sc. Computer Science · Focus: Algorithms · Graduated with honors · Grade: 1.2 (1.0 = best)

Bonn, Germany

Oct. 2022 – Sep. 2025

- Thesis (Grade: 1.0): adapted Hybrid Genetic Search (PyVRP) to the Dynamic VRPTW and outperformed three state-of-the-art ACS baselines on the majority of dynamic Solomon instances

PUBLICATIONS

Reinforcement Learning Amplifies Emergent Misalignment from Harmless Rewards

EMNLP 2026

- Showed that RL on rewards with no apparent harmful content still amplifies broad misalignment
- Implemented and evaluated mitigation techniques against RL-induced emergent misalignment

Multi-Agent Reinforcement Learning for Complex and Dynamic Graph-Based Planning Problems *ICMCIS 2026*

- Developed graph reinforcement learning with GNNs and MAPPO for dynamic multi-agent airlift planning

EXPERIENCE

Incoming AI & Operations Research Intern

DHL Group

Bonn, Germany

Oct. 2026 – Dec. 2026

- LLM agent-based query system over internal operational data and delivery-district optimization

Research Assistant

Fraunhofer Institute FKIE

Bonn, Germany

Nov. 2023 – Sep. 2026

- Trained and evaluated GNN-based RL policies for dynamic multi-agent planning problems
- Developed online combinatorial optimization methods for dynamic routing and resource allocation under uncertainty
- Contracted Germany-wide OpenStreetMap road graphs with 10M+ edges into solver-ready instances

COMPETITIONS

Google DeepMind: Measuring Progress Toward AGI

Apr. 2026

- Co-authored PDPBench, a PDPTW benchmark evaluating frontier LLMs on request insertion, route construction, and full-solution generation

ICAPS 2024: AFRL Airlift Challenge

Apr. 2024

- Placed 2nd in the ICAPS 2024 AFRL Airlift Challenge for dynamic cargo-delivery planning
- Designed the core routing approach for heterogeneous aircraft operating on dynamic, disrupted networks

HONORS

Deutschlandstipendium (German National Scholarship)

Oct. 2024, 2025 & 2026

- Merit-based federal scholarship for outstanding academic achievement

SKILLS

Programming Languages: Python, C++

Libraries & Frameworks: PyTorch, Transformers, NumPy, Pandas, Gymnasium, NetworkX

Tools: Git, Linux, Docker, Slurm, OR-Tools

Languages: German (native), English (C1), Spanish (B1)